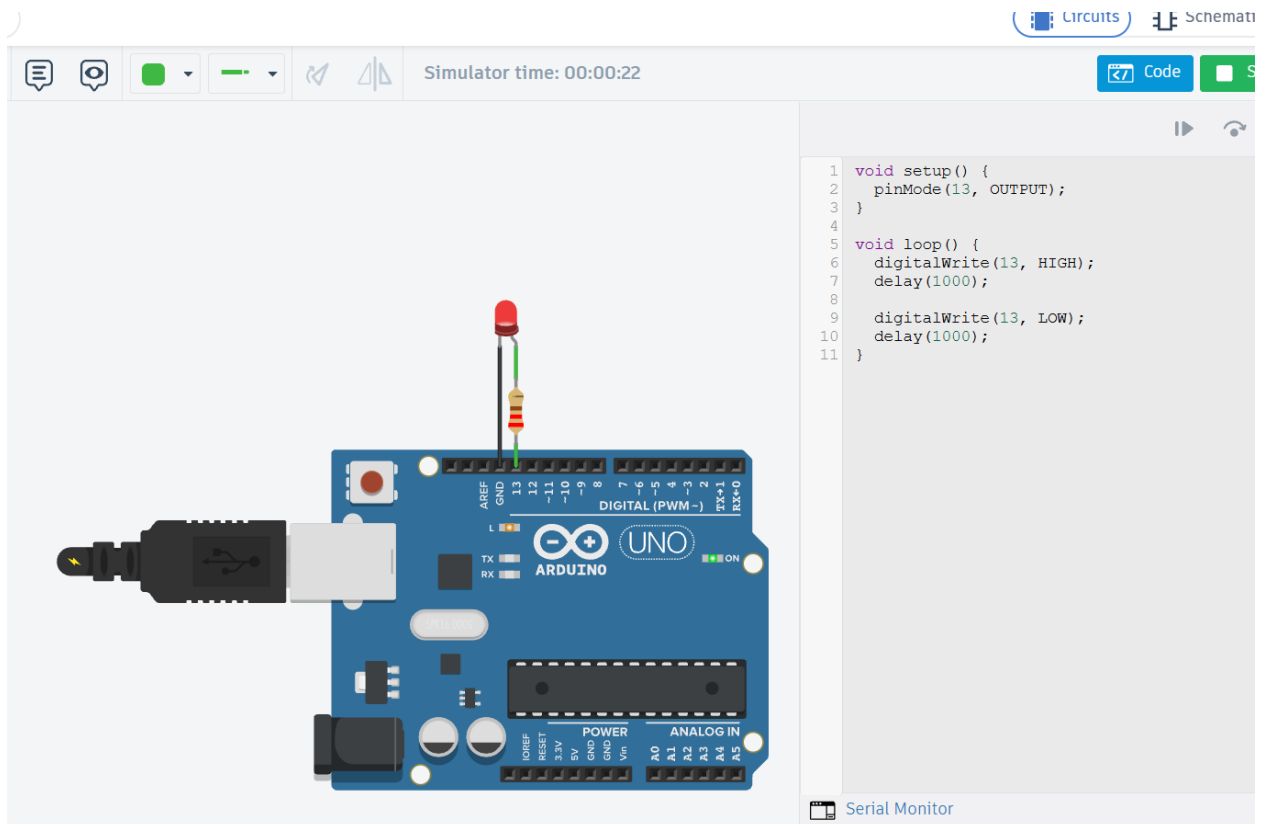


BLINK AN LED

CODE:

```
void setup() {  
  pinMode(13, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(13, HIGH);  
  delay(1000);  
  digitalWrite(13, LOW);  
  delay(1000);  
}
```



BLINK AN LED WITH BUTTON

```
int button = 2;
```

```
int led = 8;
```

```
void setup() {
```

```
    pinMode(button, INPUT_PULLUP);
```

```
    pinMode(led, OUTPUT);
```

```
}
```

```
void loop() {
```

```
    if (digitalRead(button) == LOW) {
```

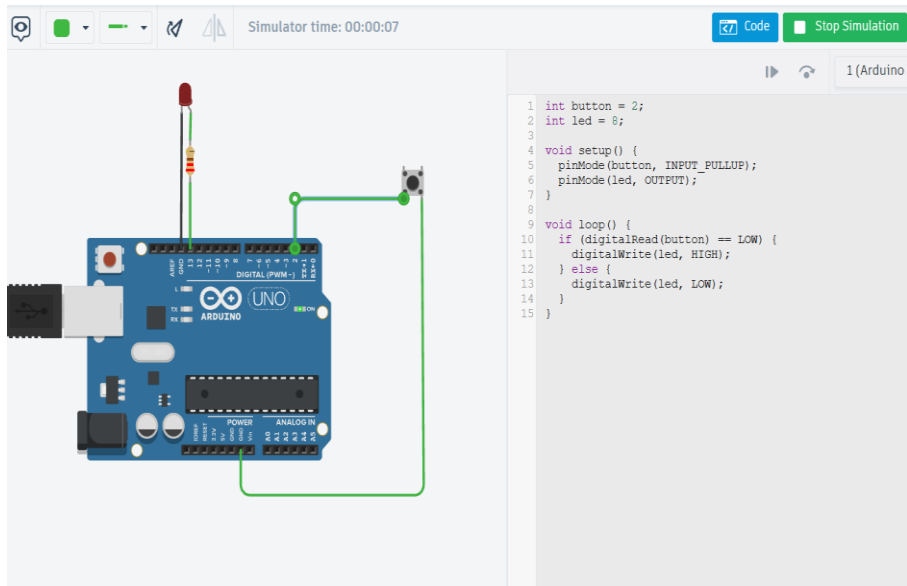
```
        digitalWrite(led, HIGH);
```

```
    } else {
```

```
        digitalWrite(led, LOW);
```

```
    }
```

```
}
```



USING AN ULTRASONIC SENSOR

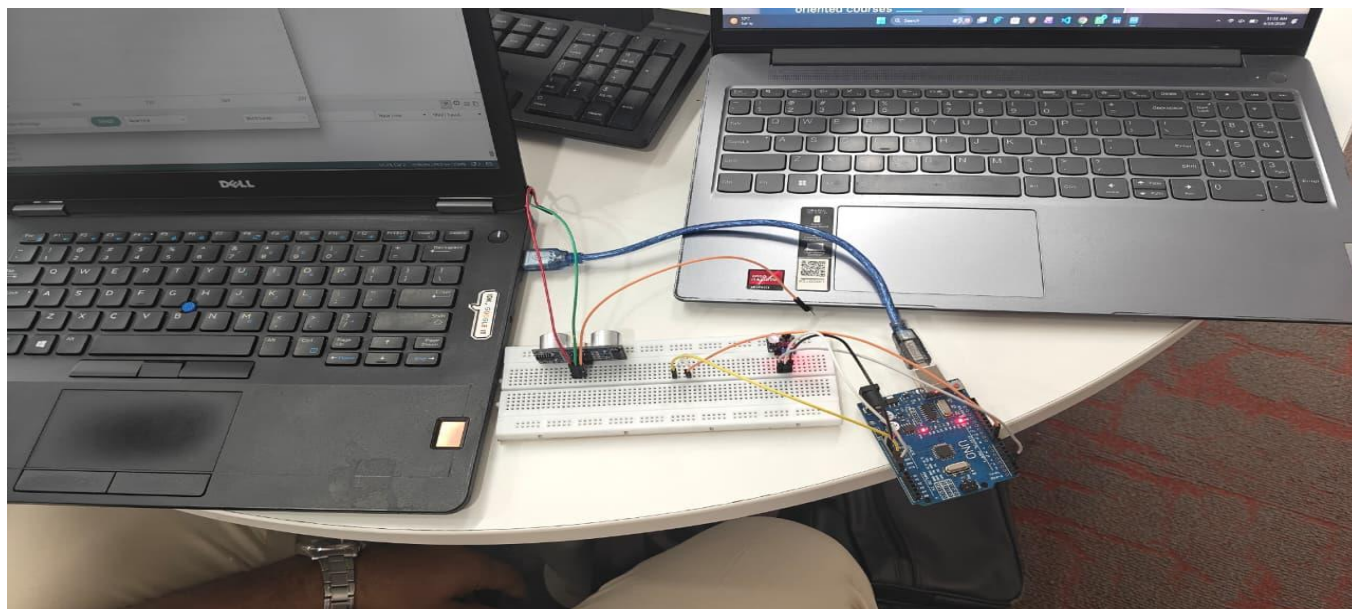
CODE:

```
#define TRIG 9
#define ECHO 10

long duration;
float distance;

void setup() {
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  duration = pulseIn(ECHO, HIGH);
  distance = duration * 0.0343 / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
  delay(500);
}
```



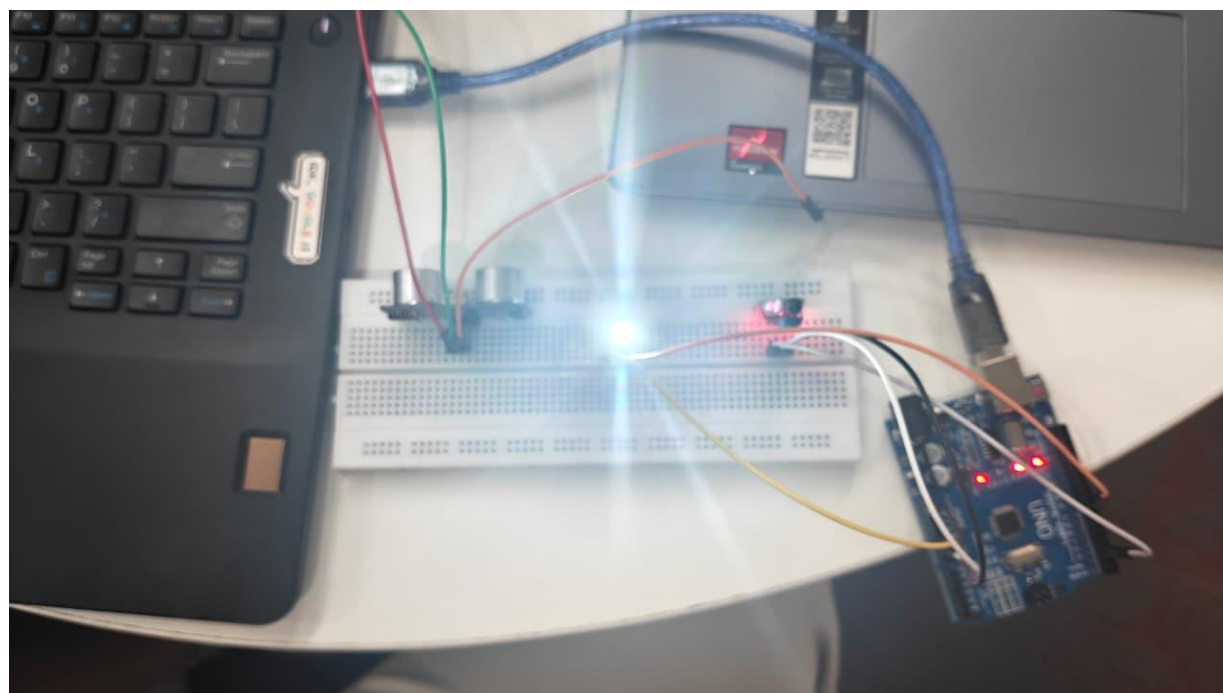
USING IR SENSOR WITH LED BLINKING TO SHOW DETECTION

CODE :

```
#define IR_PIN 2
#define LED_PIN 8

void setup() {
  pinMode(IR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int irState = digitalRead(IR_PIN);
  if (irState == LOW) {
    // Object detected
    digitalWrite(LED_PIN, HIGH);
    Serial.println("OBJECT DETECTED");
  } else {
    digitalWrite(LED_PIN, LOW);
    Serial.println("NO OBJECT");
  }
  delay(100);
}
```



USING A TEMPERATURE SENSOR

```
int sensorPin = A0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  int reading = analogRead(sensorPin);

  float voltage = reading * (5.0 / 1023.0);

  float temperatureC = (voltage - 0.5) * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperatureC);
  Serial.println(" °C");

  delay(1000);
}
```

Simulator time: 00:00:05

Code

Stop Simulation

1 (Arduinc

```
1 int sensorPin = A0;
2
3 void setup() {
4   Serial.begin(9600);
5 }
6
7 void loop() {
8   int reading = analogRead(sensorPin);
9
10  float voltage = reading * (5.0 / 1023.0);
11
12  float temperatureC = (voltage - 0.5) * 100.0;
13
14  Serial.print("Temperature: ");
15  Serial.print(temperatureC);
16  Serial.println(" °C");
17
18  delay(1000);
19 }
```

Serial Monitor

Temperature: 24.78 Å°C
Temperature: 24.78 Å°C
Temperature: 24.78 Å°C
Temperature: 24.78 Å°C
Temperature: 24.78 Å°C
Temperature: 24.78 Å°C

Text input

Send

The image shows a simulated Arduino Uno board connected to a temperature sensor (TMP) via green wires. The sensor is connected to the A0 pin. The board is connected to a USB cable. The code on the right reads the sensor value and prints the temperature in Celsius. The Serial Monitor shows the output: 'Temperature: 24.78 Å°C' repeated six times.